CHAPTER NO-1

INTRODUCTION

CHAPTER NO-1

INTRODUCTION

1.1 Introduction-

In this project we are taking coding solutions of given problems from students based on their programming language. Many times our subject teachers give assignment to their students but students did not solve the problems seriously they just roat the code but actually roating code's is not good for collage exam and interview, it needs daily practice.

But our students not practicing daily. For that problem we developed a solution called "E-learning platform" . We developed online solution because when student are in there home we don't know he solve coding problems or not. in this project there are two types of login first is teacher login who assign the question belong to their subject and second login is student login where students solve the question given by particular subject teacher.

Student can able to solve this questions and teacher can view their solutions. Student cannot copy code from any platform because we block copy paste in our platform. And other section of platform is leaderboard system. Based on student performance their rank will show. Each question have time constraint. Question must be solved in given time.it become compulsory to all students to solve coding questions.

1.2 Purpose of the project-

E-Learning, or electronic learning, is the delivery of learning and training through digital resources. Although eLearning is based on formalized learning, it is provided through electronic devices such as computers, tablets and even cellular phones that are connected to the internet. This makes it easy for users to learn anytime, anywhere, with few, if any, restrictions. Basically, eLearning is training, learning, or education delivered online through a computer or any other digital device.

One of the importance of e-learning in education is that tutors and participants both can develop advanced learning skills. For example, improve DSA Logic and problem solving logic.

This is important in current times as competition is rising and the world is also growing.

Hence, quick information helps in the better growth of an individual..

1.3 Scope of the project-

Through E-Learning, you can save time, and money and reduced costs. so, E- Learning is cost-effective compared to traditional learning.

E-Learning is provided online so No need for papers like traditional learning.

Thus, we can save the Environment through this.

As computer ownership grows across the globe e-learning becomes increasingly viable and accessible. Internet connection speeds are increasing, and with that, opportunities for more multimedia training methods arise.

CHAPTER NO - 2

LITERATURE SURVEY

CHAPTER NO - 2
LITERATURE SURVEY

2.1 Existing System

In Current System our students don't take assignment and programming language serious. because Most students find learning offline boring and often complain of lacking the motivation to make it through a class. They don't have much interest on it. If any student miss the lecture so he cannot take back the lecture. Even teachers say that the attendance of students is decreasing.

2.2 Proposed System

In this project eLearning allows students to quickly and more easily complete their assignment, resulting in improved performance and greater productivity. Students can solve the problem related to their subjects. And get motivated. And improve their self.

CHAPTER NO-3

PROJECT PLAN / ACTION PLAN

CHAPTER NO-3

PROJECT PLAN / ACTION PLAN

3.1 Project Planning-

Table No. 3.1: Project Planning

Sr.No	Activities	Duration In Week
1	Plan Of Project	01 Week
2	Designing/ Module Formation	02 Week
3	Coding	04 Week
4	Testing	02 Week
5	Final Report Writing / Documentation	03 Week

3.2 Gantt Chart-

Table No. 3.2: Project Planning

Activity Name	Month	February				March				April				May			
	Week	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Plan of Project																	
Designing and module Formation																	
Coding																	
Testing																	
Final reporting writing/Documentation																	

CHAPTER NO-4

SYSTEM REQUIREMENTS

CHAPTER NO-4

SYSTEM REQUIREMENTS

3.1 Software Requirements

Language	NodeJs
Database	MongoDB
Software development kit	Node V16.9.0

3.2 Hardware Requirements

Name of Component	Specification
Processor	I3
Ram	4 GB MIN
Storage	50GB MIN
Operating System	Windows 10

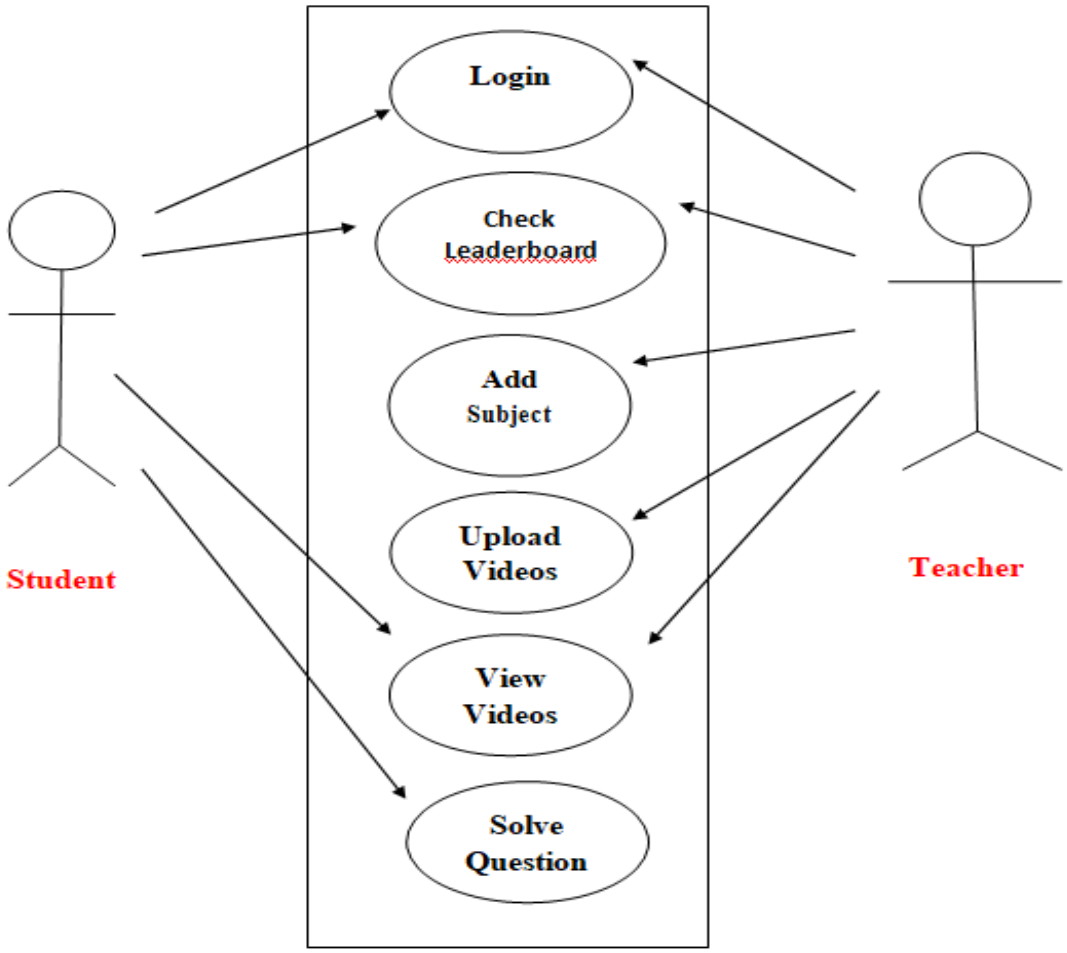
CHAPTER NO-5

SYSTEM ANALYSIS

CHAPTER NO-5

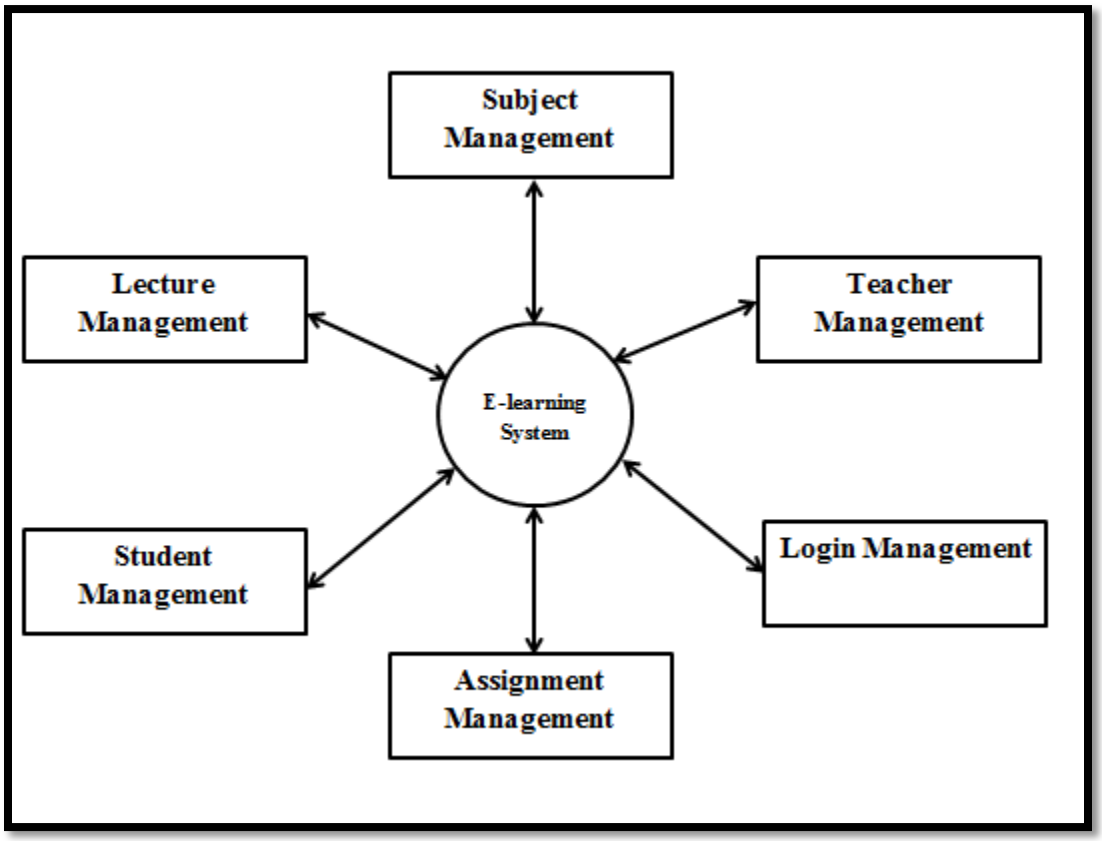
SYSTEM ANALYSIS

5.1 Use Case Diagram:-

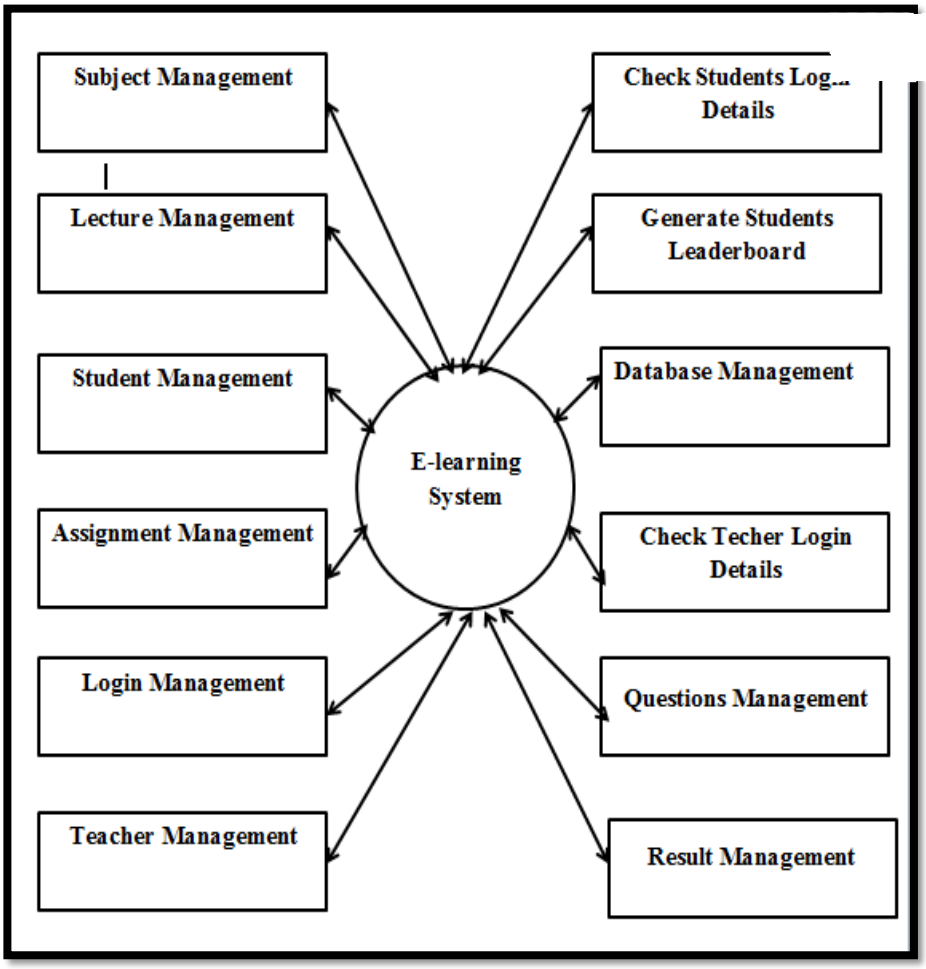


5.2 DFD Diagram:-

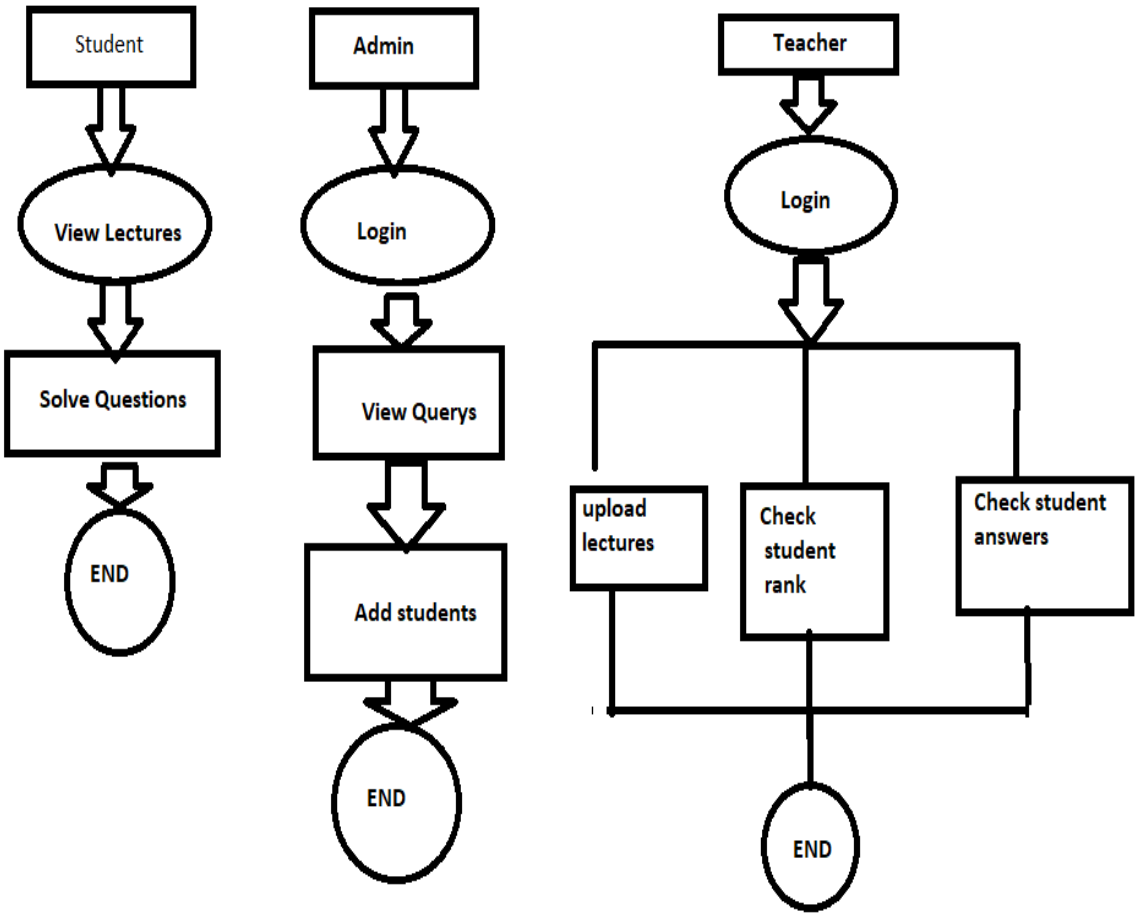
5.2.1 DFD 0 Level:-



5.2.2 DFD 1 Level:-



Activity Diagram-



CHAPTER NO-6

FEASIBILITY STUDY

CHAPTER NO-6

FEASIBILITY STUDY

4.1 Introduction to feasibility

A feasibility study aims to objectively and rationally uncover the strengths and weaknesses of an existing business or proposed venture, opportunities and threats present in the environment, the resources required to carry through, and ultimately the prospects for success. In its simplest terms, the two criteria to judge feasibility are cost required and value to be attained. A well-designed feasibility study should provide a historical background of the business or project, a description of the product or service, accounting statements, details of the operations and management, marketing research and policies, financial data, legal requirements and tax obligations. Generally, feasibility studies precede technical development and project implementation. A feasibility study evaluates the project's potential for success; therefore, perceived objectivity is an important factor in the credibility of the study for potential investors and lending institutions. It must therefore be conducted with an objective, unbiased approach to provide information upon which decisions can be based.

4.2 Technical Feasibility 2

The operational feasibility assessment focuses on the degree to which the proposed development projects fits in with the existing business environment and objectives with regard to development schedule, delivery date, corporate culture, and existing business processes. To develop this project we are using windows operating system. All operations done by using Android language. Operationally this project is feasible because it can handle the all problems related to existing system. This assessment is based on an outline design of system requirements. When writing a feasibility report, the following should be taken to consideration:

A brief description of the system to assess more possible factors which could affect the study

- The part of the current trading process being examined

- The human and economic factor
- The possible solutions to the problem

To develop our system, we required knowledge of Android. We know the Android language. There are easily available all the related software to develop this project. Java and android is easy to implement. So that, our system is technically feasible. This is platform independent.

4.3 Operational Feasibility

Operational feasibility is a measure of how well a proposed system solves the problems, and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development. Proposed is said to be operationally feasible as it provide the user friendly Graphical User Interface(GUI). User will able to understand the system clearly and correctly and can use the system with ease.User do not need any special training for operating this application. This application will be designed more user friendly as considering from the point of view of bike users.

4.4 Economical Feasibility

The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide. It includes quantification and identification of all the benefits expected. This assessment typically involves a cost/ benefits analysis. We know that, Android implementations are open source. We can get all the things related to android from internet. All the software required to develop this project are available on internet. So that, we can easily get these software to develop our system. We did not require buying any software for this project. So that, our system is economically feasible.

4.5 Legal Feasibility

Determines whether the proposed system conflicts with legal requirements. We are developing this project for academic purpose and for solving the problems faced by peoples and bike users. This project will be useful for bike users. So that, our system is legally feasible.

CHAPTER NO -7

TECHNICAL OVREVIEW

CHAPTER NO -7

TECHNICAL OVREVIEW

Following are some languages used while developing the website:

1. Nodejs
2. EJS
3. Mongodb

1) Nodejs

- Node.js is a server-side JavaScript runtime environment that allows developers to run JavaScript code outside of a web browser.

- It is built on the Google V8 JavaScript engine, which provides high performance and allows for efficient execution of JavaScript code.

- Node.js has an event-driven, non-blocking I/O model, which makes it ideal for building scalable network applications.

- It has a large and active community of developers who have contributed to its ecosystem with a vast number of libraries and tools.

- Node.js can be used to build a wide range of applications, including web servers, command-line tools, desktop applications, and even IoT devices.

- It supports various operating systems, including Windows, Linux, and macOS, and can be easily deployed on cloud platforms like AWS, Google Cloud, and Microsoft Azure.

- Node.js provides a set of built-in modules, such as HTTP, HTTPS, and File System, which simplifies the process of building applications.

- Node.js is often used in combination with popular front-end frameworks like React and Angular, to build full-stack web applications.

- It has a low barrier to entry and is easy to learn for developers with a background in JavaScript.

- Node.js has a strong focus on performance and scalability, making it a popular choice for high-traffic websites and applications.

2) EJS

- EJS is a templating language that allows developers to generate HTML markup with JavaScript.

- It is easy to learn and use, and can be used in a variety of web frameworks, including Node.js and Express.

- EJS allows developers to include variables, loops, conditionals, and other JavaScript constructs in their HTML markup, making it easy to generate dynamic content.

- It supports partials and layouts, which allows developers to reuse and organize code across multiple pages.

3) MongoDB

- MongoDB is a NoSQL document-oriented database that stores data in JSON-like documents with dynamic schema, allowing for flexible and scalable data modeling.

- It is open-source software and is maintained by MongoDB Inc.

- MongoDB has a scalable architecture and can be used to handle large amounts of data with high performance.

- It provides a rich query language and supports various data types, including arrays, embedded documents, and geospatial data.

- MongoDB provides high availability and fault tolerance through replica sets and sharding, which distribute data across multiple servers.

- It has a flexible data model that allows for easy updates and modifications without needing to follow a rigid schema.

- MongoDB has a large and active community of developers who have contributed to its ecosystem with a vast number of libraries and tools.

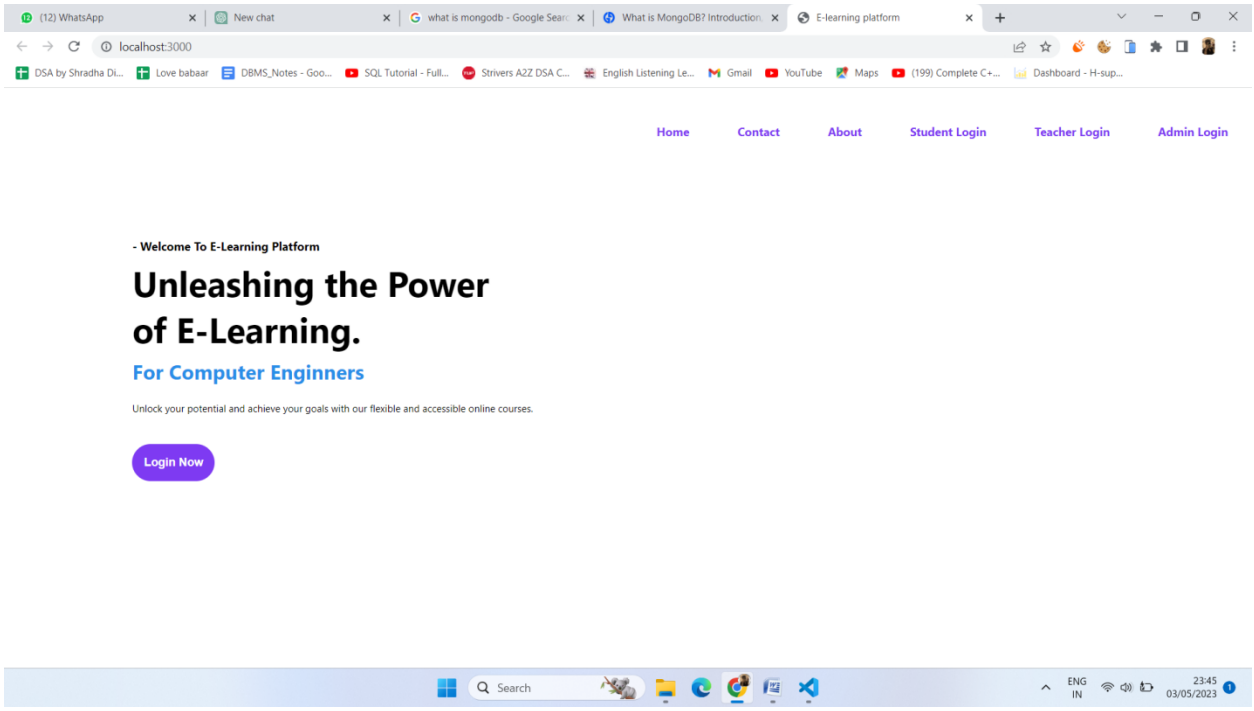
- It can be used in various applications, including web and mobile applications, IoT devices, and real-time analytics.

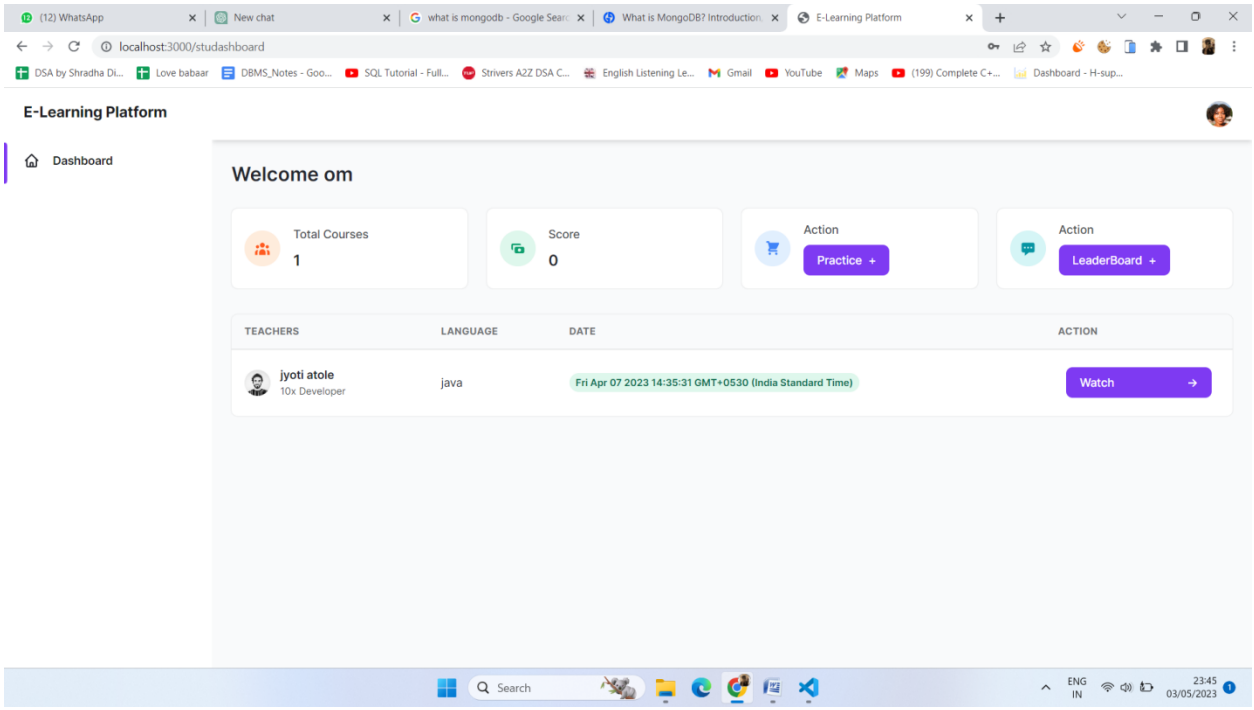
CHAPTER NO-8

SYSTEM DESIGN AND SNAPSHOTS

CHAPTER NO-8

SYSTEM DESIGN AND SNAPSHOTS





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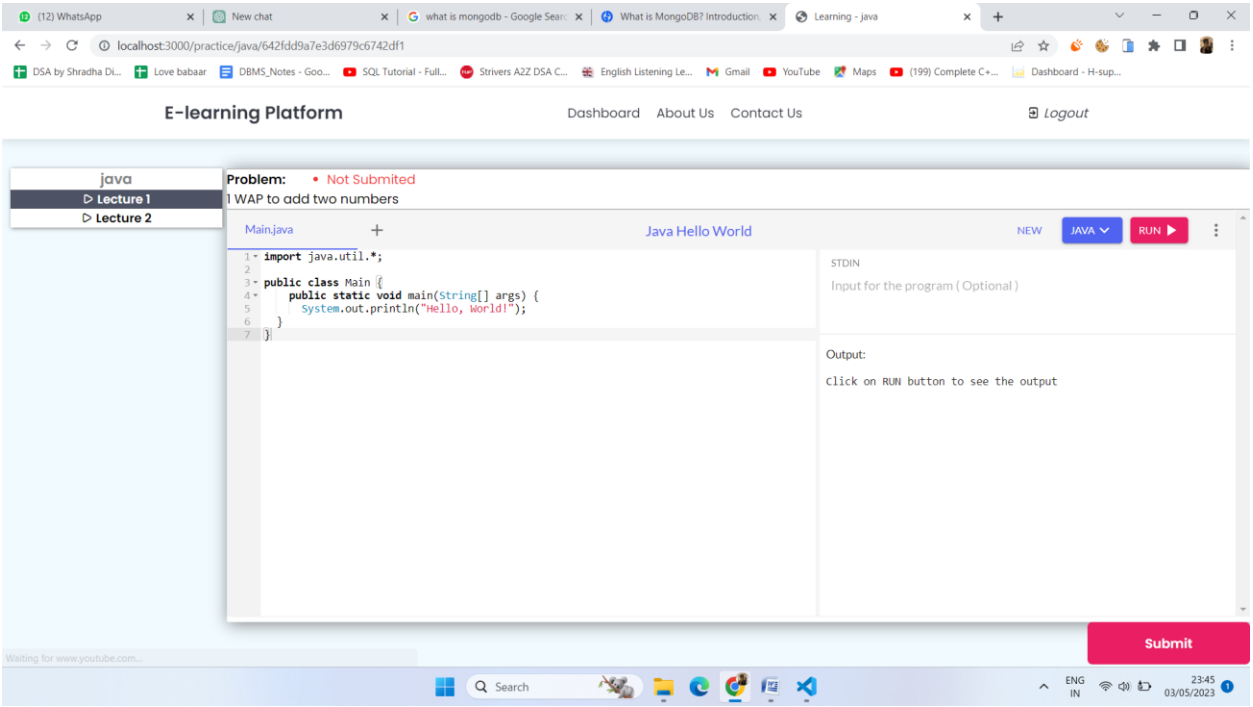
E-learning Platform
Dashboard
About Us
Contact Us
Logout
java
Introduction to Java Language | Lecture 1
[2] java for loops
Introduction to Java Language | Lecture 1 | Complete Pla...
Watch later
Share
Watch on YouTube
Microsoft
WAP to add two numbers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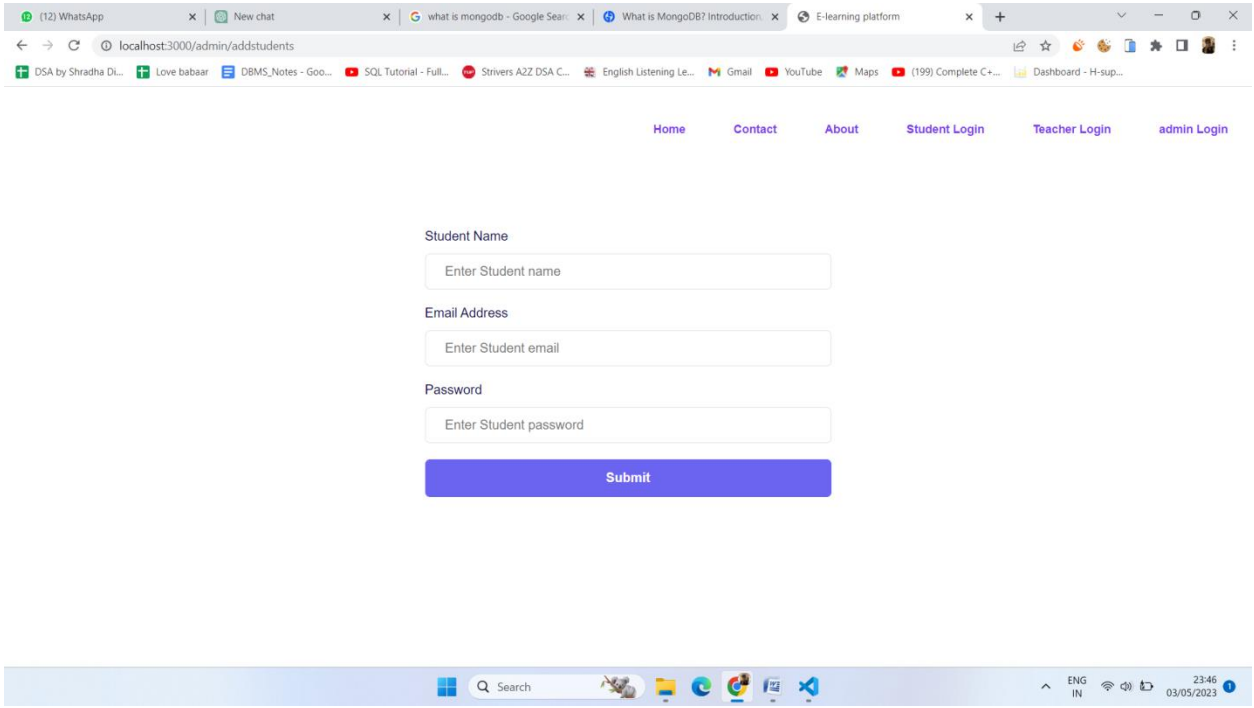
E-learning Platform
Dashboard
About Us
Contact Us
Logout
java
Lecture 1
Lecture 2
Problem:
Not Submited
1 WAP to add two numbers
Main.java
Java Hello World
NEW
JAVA
RUN
import java.util.*;
public class Main {
public static void main(String[] args) {
System.out.println("Hello, World!");
}
}
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
Submit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About
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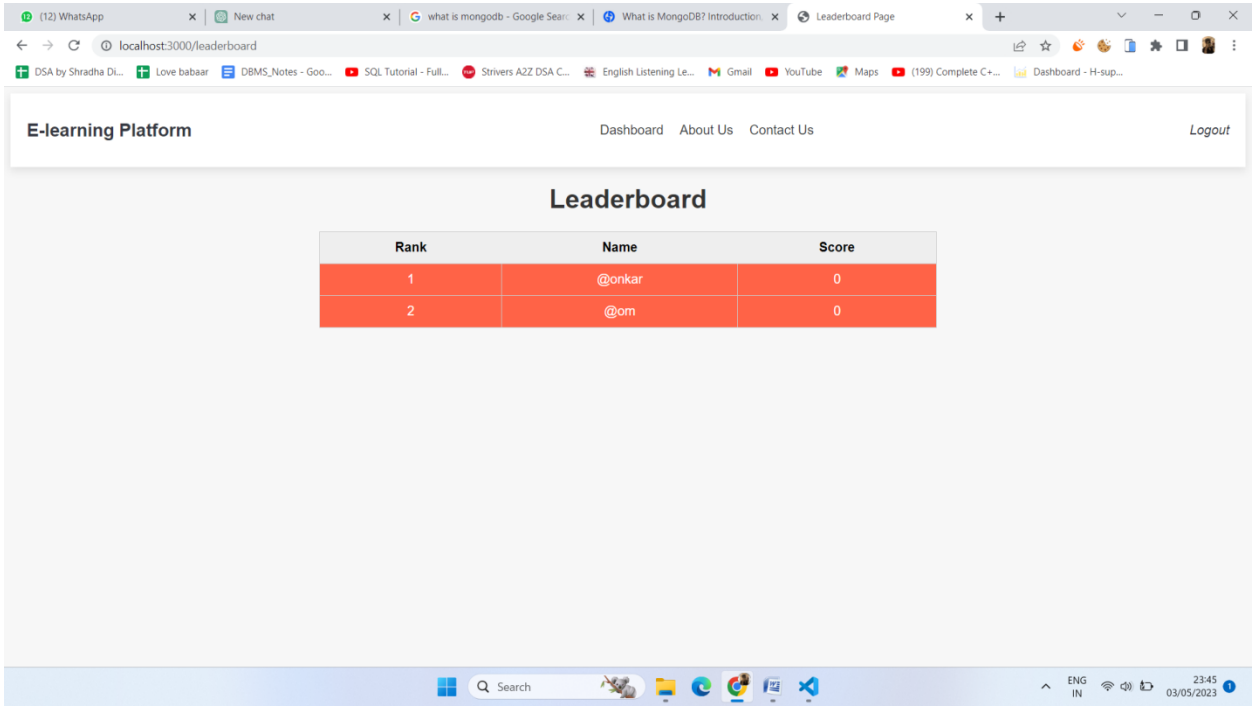
Teacher Login
admin Login
Student Name
Enter Student name
Email Address
Enter Student email
Password
Enter Student password
Submit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Score
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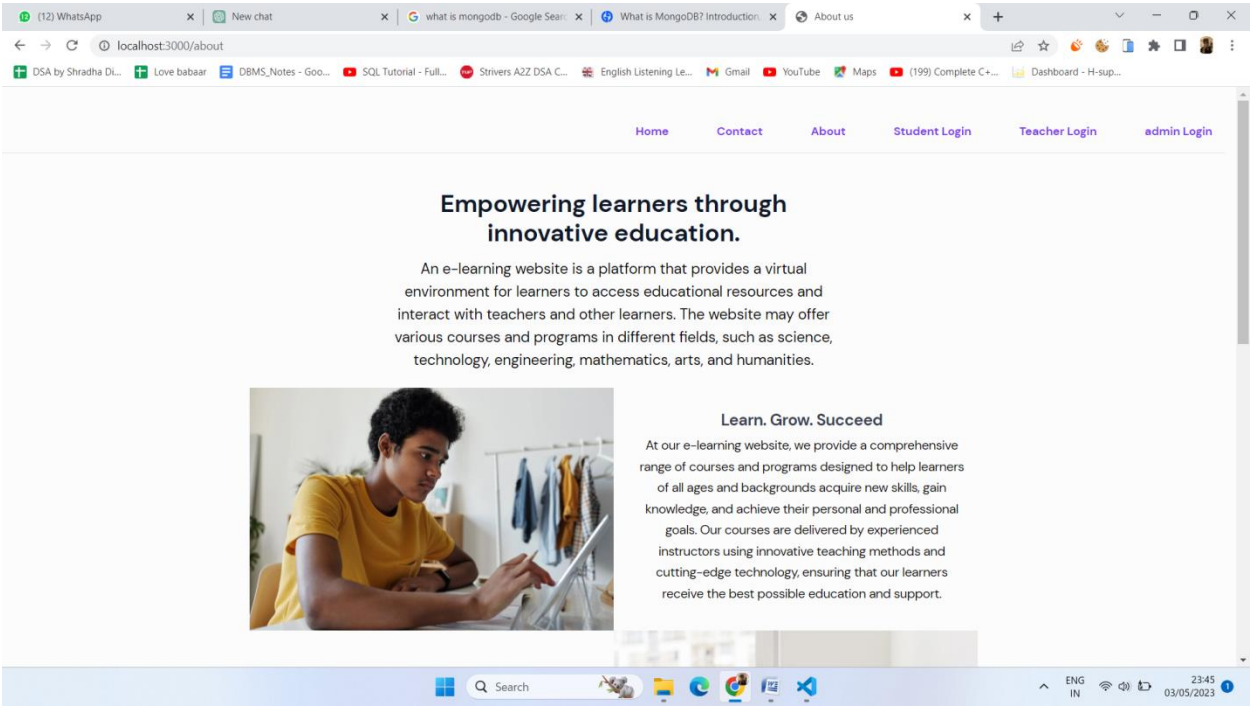
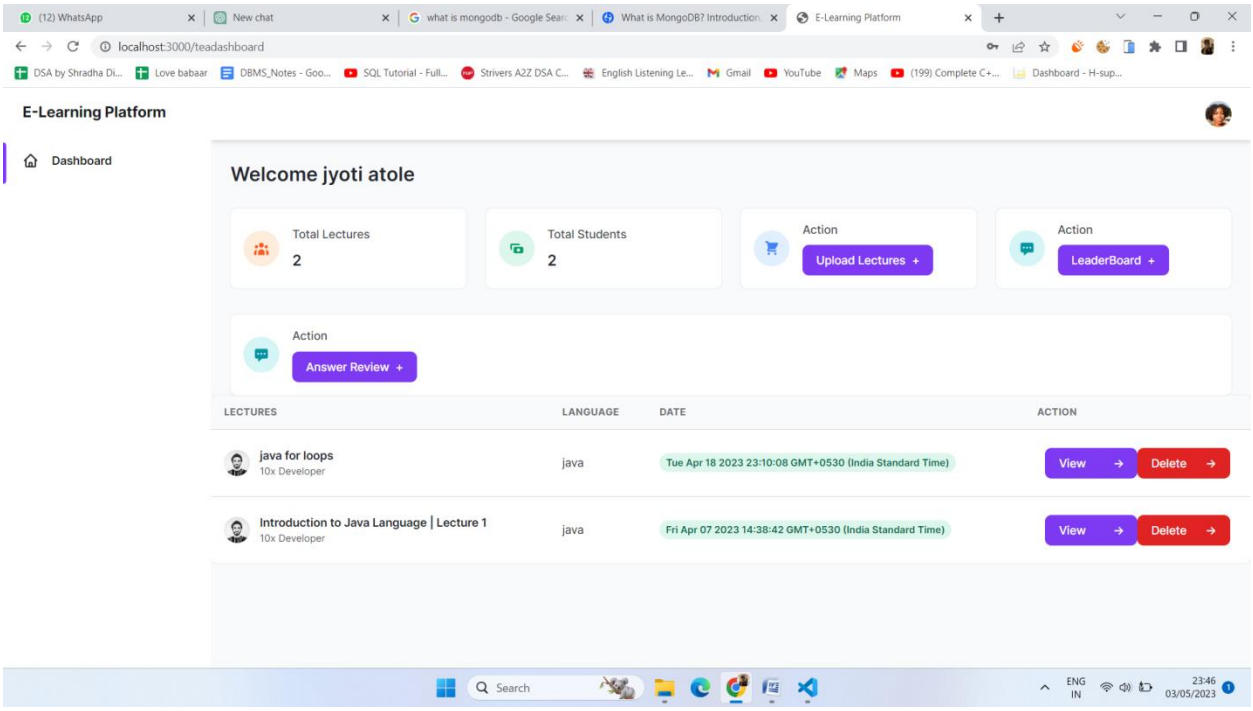

localhost:3000/about
Home
Contact
About
Student Login
Teacher Login
admin Login
Empowering learners through innovative education.
An e-learning website is a platform that provides a virtual environment for learners to access educational resources and interact with teachers and other learners. The website may offer various courses and programs in different fields, such as science, technology, engineering, mathematics, arts, and humanities.
Learn. Grow. Succeed
At our e-learning website, we provide a comprehensive range of courses and programs designed to help learners of all ages and backgrounds acquire new skills, gain knowledge, and achieve their personal and professional goals. Our courses are delivered by experienced instructors using innovative teaching methods and cutting-edge technology, ensuring that our learners receive the best possible education and support.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Total Students
2
Action
Upload Lectures +
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LeaderBoard +
Action
Answer Review +
LECTURES
LANGUAGE
DATE
ACTION
java for loops
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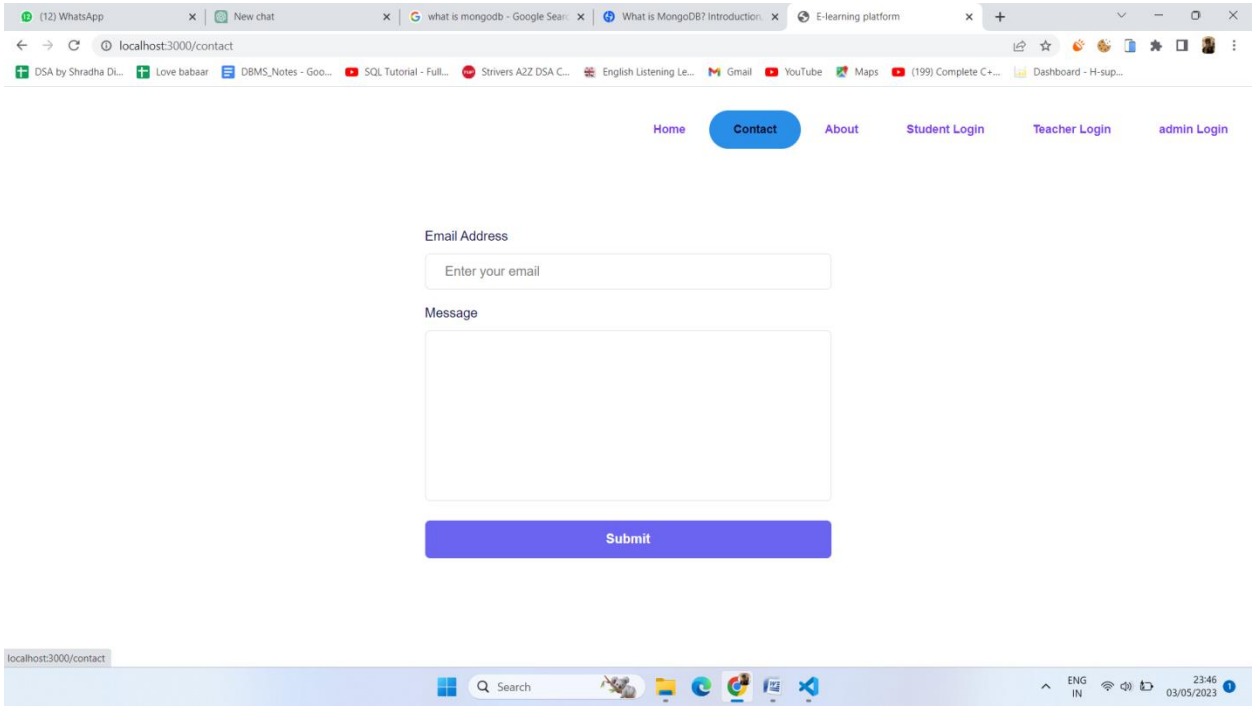
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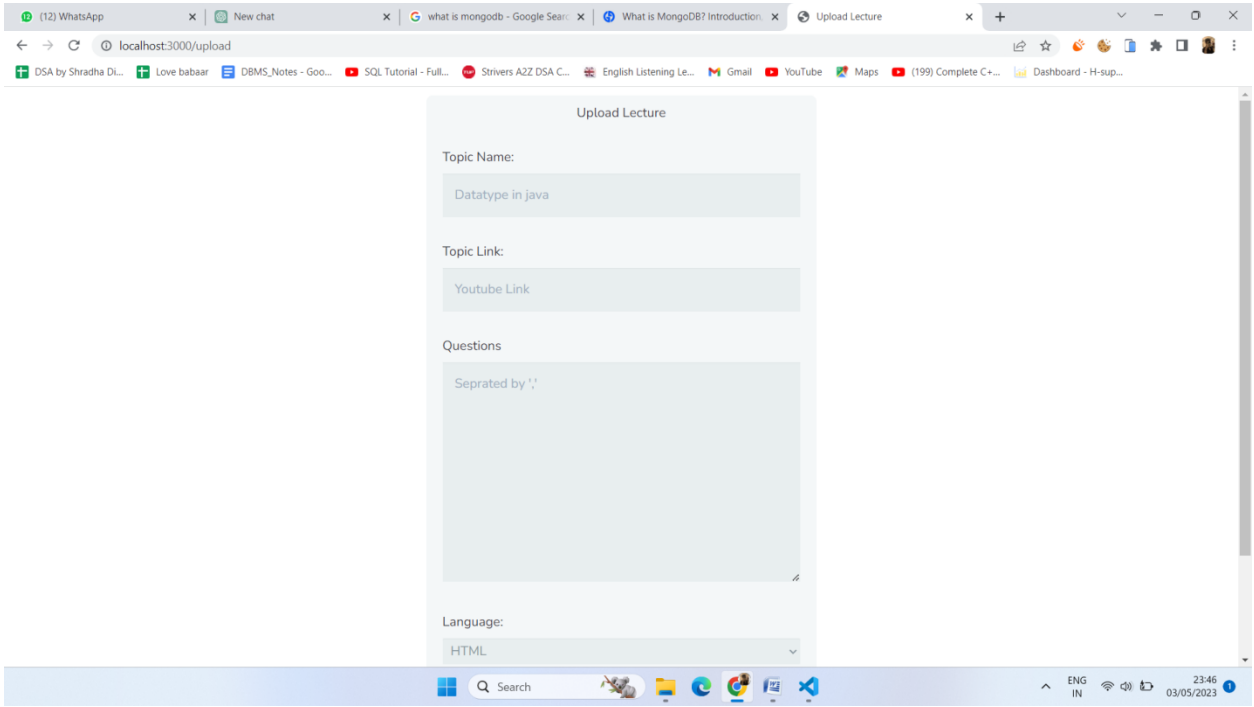
About
Student Login
Teacher Login
admin Login
Email Address
Enter your email
Message
Submit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Youtube Link
Questions
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Language:
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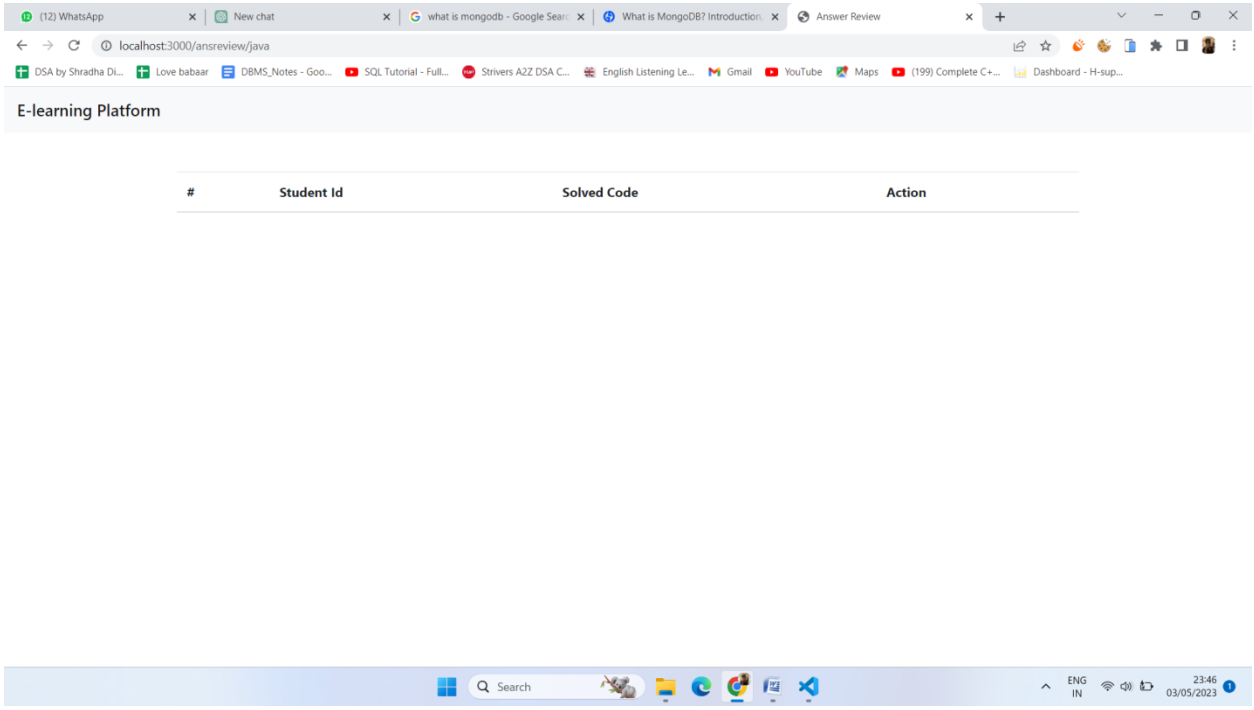
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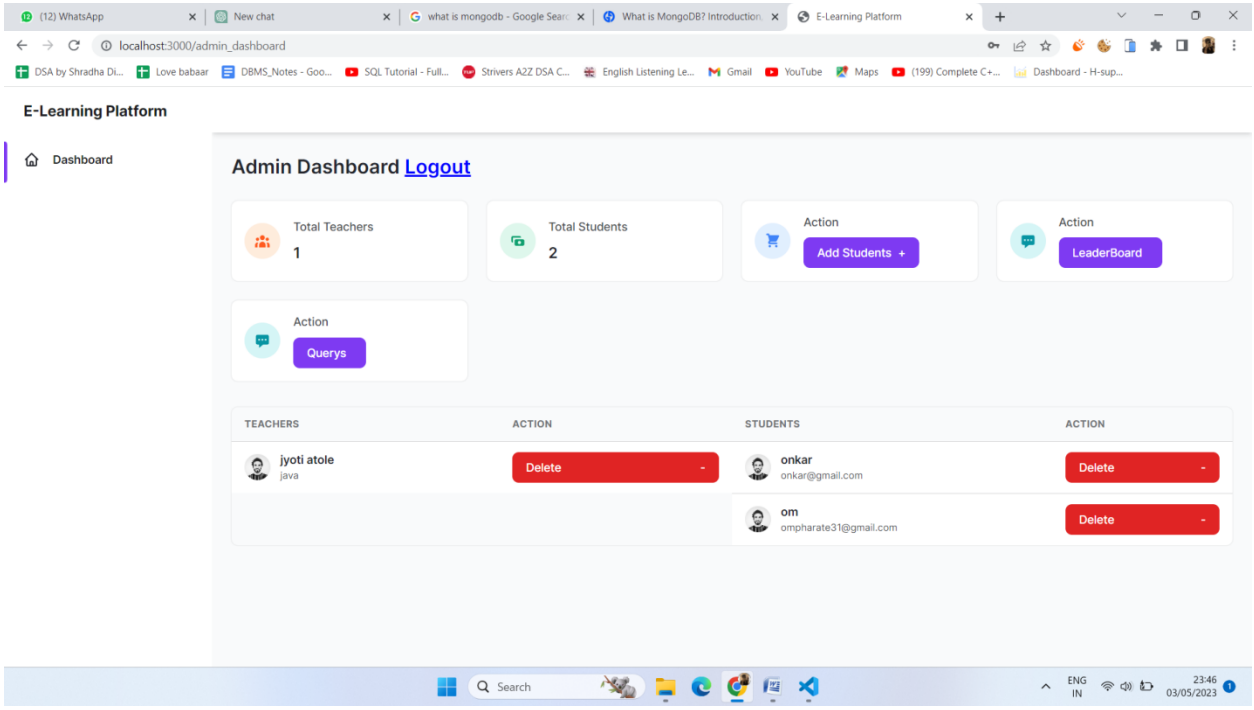
Student Id
Solved Code
Action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Dashboard
Admin Dashboard Logout
Total Teachers
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Total Students
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Action
Add Students +
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Querys
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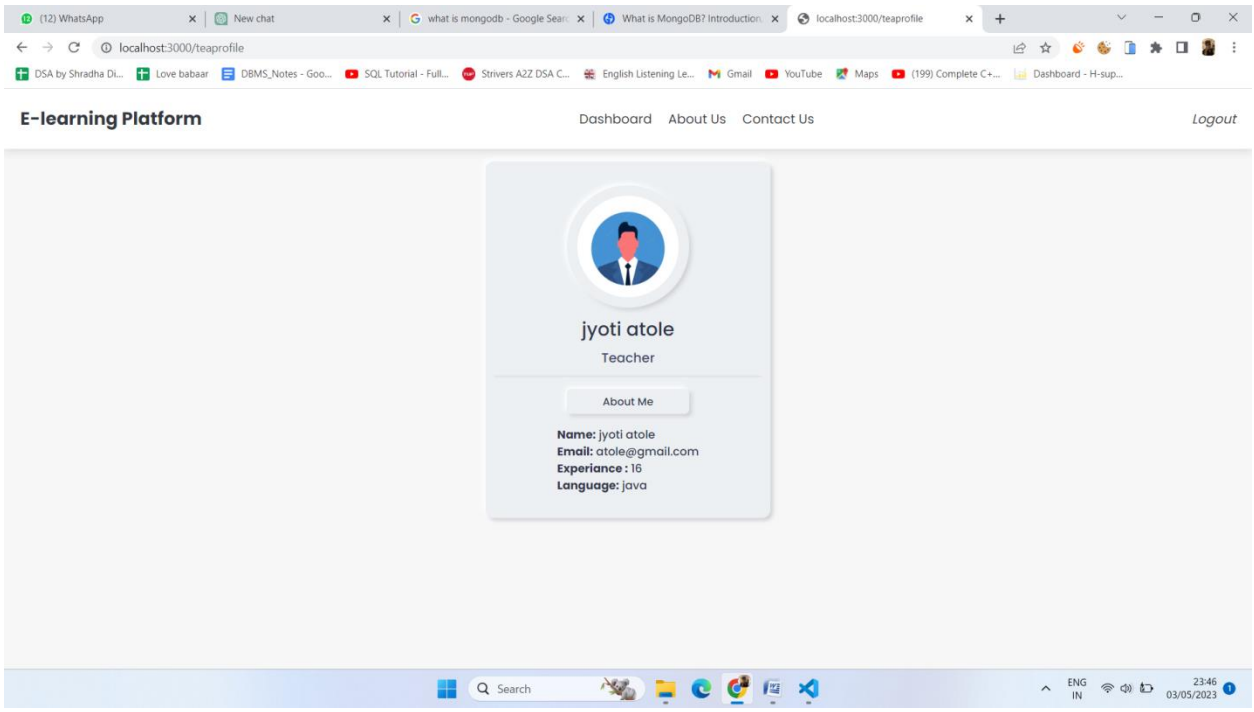
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Dashboard
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Experiance : 16
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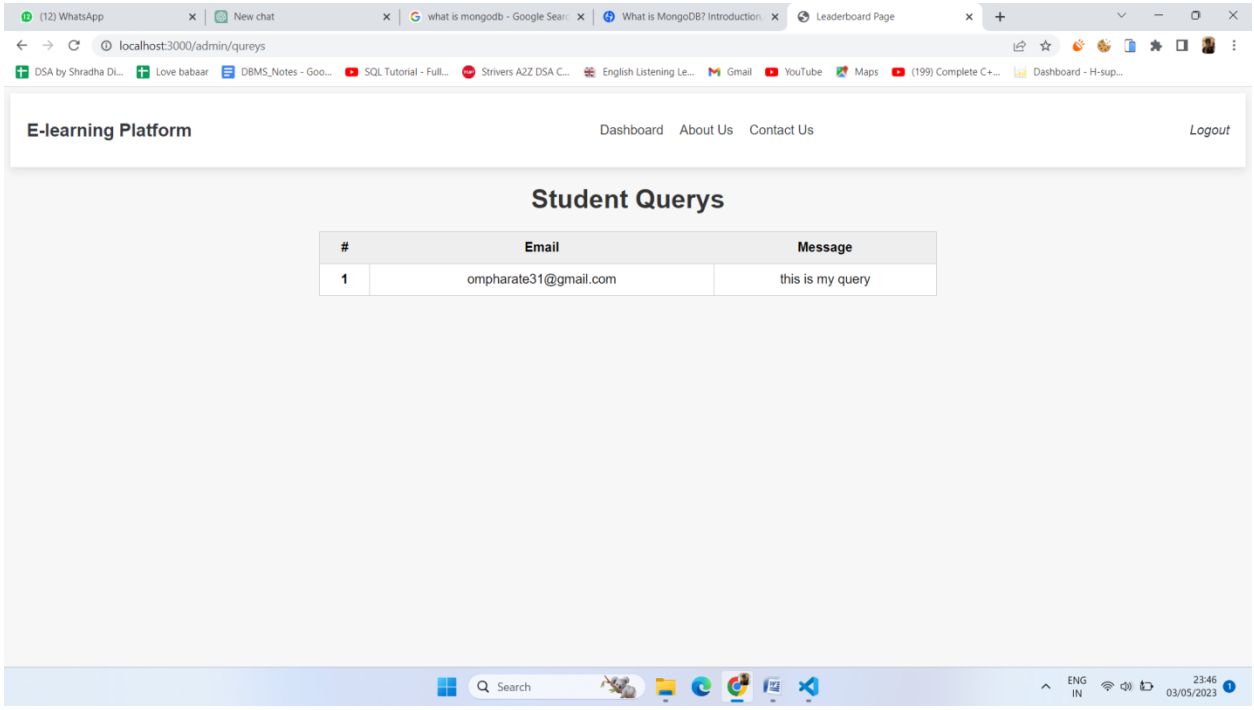
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E-learning Platform
Dashboard
About Us
Contact Us
Logout
Student Querys
| Email | Message
1 | ompharate31@gmail.com | this is my query

CHAPTER NO-9

TESTING PLAN

9.1 Basic of testing-

Software testing can be stated as the process of verifying and validating that a software or application is bug free, meets the technical requirements as guided by its design and development and meets the user requirements effectively and efficiently with handling all the exceptional and boundary cases.

The process of software testing aims not only at finding faults in the existing software but also at finding measures to improve the software in terms of efficiency, accuracy and usability. It mainly aims at measuring specification, functionality and performance of a software program or application

Step	**Step Details**	**Expected Results**	**Actual Results**	**Pass/Fail**
1	**1)** Go to the website's login page **2)** Enter valid login credentials **3)** Click on the login button	Student should be redirected to their dashboard	Student is redirected to their dashboard	**Pass**
2	**1)** Go to the course page **2)** Select the lecture to view **3)** Click on the play button	Lecture video should start playing	Lecture video starts playing	**Pass**
3	**1)** Go to the leaderboard page **2)** Verify the ranking of the student	Student's ranking should be displayed on the leaderboard	Student's ranking is displayed on the leaderboard	**Pass**
4	1). Go to the website's login page 2). Enter valid login credentials	Teacher should be redirected to	Teacher is redirected to	**Pass**

	3). Click on the login button	their dashboard	their dashboard	
5	1. Go to the course page 2. Click on the upload lecture button 3. Upload a valid lecture file **4.** Click on the upload button	Lecture should be uploaded successfully	Lecture is uploaded successfully	**Pass**
6	1. Go to the leaderboard page **2.** Verify the ranking of the students	Rankings of the students should be displayed on the leaderboard	Rankings of the students are displayed on the leaderboard	**Pass**
7	1. Go to the registration page 2. Enter valid registration details **3.** Click on the register button	Teacher should be registered successfully	Teacher is registered successfully	**Pass**

CHAPTER NO-10

APPLICATION AND ADVANTAGES

CHAPTER NO-10

APPLICATION AND ADVANTAGE

10.1Applications of E-learning website.

1. Education and Training: E-learning websites can be used to provide education and training in various fields such as academic subjects, technical skills, and professional development. These websites can provide access to courses, lectures, assignments, quizzes, and exams, allowing learners to gain knowledge and skills from anywhere with an internet connection.

2. Corporate Training: E-learning websites can also be used by businesses and organizations to provide training to their employees. This can include courses on job-specific skills, workplace safety, and compliance training. E-learning can be a cost-effective solution for businesses since it eliminates the need for in-person training sessions and can be accessed remotely.

3. Continuing Education: E-learning websites can be used to provide continuing education and professional development courses for adults who want to improve their skills or stay up-to-date with industry trends. This can include courses on finance, healthcare, law, and other industries.

4. Personal Development: E-learning websites can also be used for personal development and self-improvement. These websites can provide courses on topics such

10.2Advantages of E-learning website.

- Flexibility
- Cost-effective
- Convenience
- Access to a wider range of courses and resources
- Self-paced learning
- Personalized learning experience

CHAPTER NO-11

FUTUREENHANCEMENT

Here are some future enhancements of our E-learning project website:

1. Integration of interactive and immersive learning technologies such as virtual and augmented reality to enhance the learning experience.

2. Personalized learning paths based on the learner's interests, strengths, and weaknesses, which could be achieved through the use of AI and machine learning algorithms.

3. Real-time collaboration and communication tools, such as video conferencing and instant messaging, to enable learners to work together and learn from each other.

4. Gamification features that incorporate game elements such as badges, leaderboards, and rewards to motivate learners and make learning more engaging.

5. Adaptive testing that dynamically adjusts the difficulty of assessment based on the learner's performance and mastery of content.

6. Integration of social learning tools to enable learners to connect with others in their field and share knowledge and ideas.

10. Integration of real-world projects and internships that allow learners to apply their skills and knowledge to real-world problems and gain practical experience.

CHAPTER NO-12

COST OF PROJECT

Chapter-12

COST OF PROJECT

LOCBasedEstimation:

Table No. 12.1: line of code estimation

Sr. No.	Software Module	LOC
1.	Nodejs	1020
2.	EJS	2300
Total		3320

We calculate the project cost on the basis of No of People, No of working days and System requirement for software products.

Month required for Project development =**3 Months/90 days**

Total Number of members in project team = **03**

Total Estimated LOC = **3320**

Using COCOMO (Constructive Cost Model) Model:

Given

Ab = 2.4

Bb = 1.05

Cb = 2.5

Db = 0.38

(Coefficient factors for Semidetached Project catageory)

Efforts Estimation[E] = [Ab*(KLOC)^Bb]

= [2.4*(3.32)^1.05]

= 8.46

= 8 person/month

Duration[D] = Cb(E)^Db

= 2.5(8)^0.38

= 5.5 months

= 5

Person Estimation[p] = E/D

= 8/5

= 1.6

= 2 person

Cost Calculation: As per the we assume that the software engineering cost for each100 LOC is $10, therefore overall cost to develop the project is

= (Total LOC /100)*rate

= (3320/100)*10

= 33.2*10

= $332

CHAPTER NO-13

CONCLUSION

CONCLUSION

Online learning is beneficial to the students, tutors and the institution offering these courses. I would therefore recommend that online learning be implemented on all learning institutions and research on how to improve this learning process should be carried out.

CHAPTER NO-14

BIBLIOGRAPHY

CHAPTER NO-14

BIBLIOGRAPHY

Reference books and websites used during the entire project……..

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2) Nodejs	- David Herron,
3) NoSQl - MongoDB	- Gaurav Vaish

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